STEERING BUTTON BOX

ESP32 DevKit (CP2102 USB) — Wiring Guide | Short reference for the finalized code

1. USB / CP2102 connection

Agar aapke ESP32 DevKit board par CP2102 chip built-in hai, programming aur power ke liye board ko USB data cable se PC ke saath connect karein. Buttons/potentiometers ko CP2102 chip ke pins par nahi, ESP32 ke labeled GPIO pins par connect karna hai. Alag CP2102 module ki zaroorat nahi hoti.

2. Button wiring (toggle switches GPIO 5 and GPIO 18 are intentionally left out)

ESP32 GPIO	Button ki doosri wire	Logic
GPIO 32	3.3V	Press = HIGH
GPIO 25	3.3V	Press = HIGH
GPIO 26	GND	Press = LOW
GPIO 33	GND	Press = LOW
GPIO 4	GND	Press = LOW
GPIO 21	GND	Press = LOW
GPIO 22	GND	Press = LOW
GPIO 23	GND	Press = LOW
GPIO 19	GND	Press = LOW
GPIO 15	GND	Press = LOW

Har button ki ek wire uske listed GPIO pin par lagegi. GPIO 32 aur GPIO 25 wale buttons ki doosri wire 3.3V par; baqi listed buttons ki doosri wire GND par lagegi. Code mein in pins ke liye internal pull-down/pull-up set hai.

3. Potentiometer wiring (4 pots)

Potentiometer	Outer pin 1	Middle pin (signal)	Outer pin 2
Pot 1 — X axis	3.3V	GPIO 34	GND
Pot 2 — Y axis	3.3V	GPIO 13	GND
Pot 3 — Z axis	3.3V	GPIO 14	GND
Pot 4 — RX axis	3.3V	GPIO 27	GND

Potentiometer ke 3 pins mein se dono side wale pins supply hain aur beech wala signal hota hai. Agar direction ulta ho to outer 3.3V aur GND wires aapas mein swap ki ja sakti hain; middle signal wire apne GPIO par hi rahe.

4. Important safety checks

- ESP32 GPIO par 5V kabhi na lagayen — sirf 3.3V logic use karein.
- 3.3V aur GND ko seedha short na karein.
- GPIO 5 aur GPIO 18 toggle switches is sheet mein include nahi kiye gaye, jaisa aapne kaha.
- Button mapping / potentiometer axis mapping isi finalized code ke mutabiq hai.

Quick reference

Buttons: GPIO 32, 25 → button to 3.3V; GPIO 26, 33, 4, 21, 22, 23, 19, 15 → button to GND.
Pots: GPIO 34, 13, 14, 27 are signal inputs; all pots get 3.3V and GND on their outer pins.